Philippine Journal of Science
152 (1): 485-499, February 2023
ISSN 0031 - 7683
Date Received: 28 Jun 2022

Evaluation of Geochemical Signature in Soil Sampled from a 2004 Indian Ocean Tsunami-stricken Region in Aceh Province Located in the Western Part of Indonesia Using Scanning Electron Microscopy–Energy Dispersive X-ray (SEM-EDX) Spectroscopy and Its Compatibility with X-ray Fluorescence (XRF) Measurement

R. Mitaphonna[1], M. Ramli[2], N. Ismail[3,4], and N. Idris[3,4*]

[1]Graduate School of Mathematics and Applied Sciences,
Universitas Syiah Kuala, Darussalam, Banda Aceh 23111 Indonesia
[2]Department of Chemistry, Faculty of Mathematics and Natural Sciences,
Universitas Syiah Kuala, Darussalam, Banda Aceh 23111 Indonesia
[3]Department of Physics, Faculty of Mathematics and Natural Sciences,
Universitas Syiah Kuala, Darussalam, Banda Aceh 23111 Indonesia
[4]Department of Geophysical Engineering, Faculty of Engineering,
Universitas Syiah Kuala, Darussalam, Banda Aceh 23111 Indonesia

Chemical elements in soil samples collected from the area struck by the massive Indian Ocean tsunami on 26 Dec 2004, were identified using scanning electron microscopy–energy dispersive X-ray spectroscopy (SEM-EDX) to find out potential candidates for the geochemical signature of the 2004 Indian Ocean giant tsunami. All tsunami-affected soils were collected from three areas in Aceh Province, Indonesia, the worst stricken region by the Indian Ocean tsunami in 2004. SEM analysis showed that the damaged soil samples exhibited smaller particles and an inhomogeneous particle size distribution, which could be attributed to dissolved chemical element sedimentation after flooding by tsunami saltwater. Moreover, our findings revealed that the tsunami-affected soil samples contain many chemical elements – including marine signatures (Na, Ca, Mg, K, Al, and Si), terrestrial sources (Ti and Fe), and organic components (C and O). Organic elements detected as oxygen (O) dominated those impacted soils from the Aceh region, accounting for more than half of all samples, followed by carbon (C) and silica (Si). Furthermore, the concentration ratio of a series of elements in those impacted soils – namely, Si/Ti, Si/K, and Fe/Ti – demonstrates values that are exclusively different among the various elements evaluated in this work. The distinguishing difference in the concentration ratios of the elements matches well with X-ray fluorescence (XRF) measurement results. Therefore, the concentration ratios of the elements are good geochemical signatures of marine inundation for identifying the 2004 Indian Ocean tsunami soil samples in Aceh Province, Indonesia.

Keywords: 2004 Indian ocean tsunami, tsunami-stricken soil, surface morphology, geochemical signature, elemental concentration ratio

*Corresponding author: nasrullah.idris@unsyiah.ac.id

INTRODUCTION

Deadly tsunamis triggered by subduction-zone earthquakes such as the 2004 Indian Ocean tsunami and the 2011 Tohoku tsunami have had devastating consequences on human life (Paris *et al*. 2020), and serious damage to the coastal system (Watanabe *et al.* 2022). Unfortunately, however, the enormity of the impact of these immense natural disasters on human life and the environment escaped the attention of coastal communities and disaster researchers until the tsunami occurred again (Rubin *et al*. 2017). Many scientific efforts in geodesy and seismology have provided information on earthquake rupture patterns, and records of damage caused by modern and prehistoric tsunamis; however, specific knowledge about tsunamis is still not sufficient for mitigating massive natural disasters. Specific knowledge about the frequency and recurrence of the massive tsunami that hit is crucial for disaster mitigation to minimize vulnerability and optimize the preparedness of people who live in coastal communities (Rubin *et al*. 2017).

Several scientific research on tsunami issues has provided indicators for tsunami deposits and proof from stratigraphy, geomorphology, sedimentology, paleontology, and geochronology (Pilarczyk *et al*. 2021). However, these characteristics are still difficult to apply in identifying tsunami inundation boundaries from deposits that are not visible in the geological layer (Shinozaki *et al*. 2021). Approaching the flood boundary, the grain size of tsunami deposits tends to be fine due to the weakening of the tsunami waves, which could not transport amounts of coarse-sized particles, thus making it difficult to distinguish by visual observation (Shinozaki *et al*. 2021). Against this background, the geochemical distinction between modern and prehistoric tsunamis is a prospective and susceptive indicator (Moreira *et al.* 2017; Watanabe *et al.* 2020) for discriminating between marine and terrestrial deposits (Chagué-Goff *et al*. 2017; Paris *et al*. 2021), especially if other proxies are not present (Shinozaki *et al*. 2022). Geochemical evidence can be maintained for hundreds of years (Chagué-Goff *et al.* 2018) and can be investigated from a non-tsunami deposit layer (Shinozaki *et al.* 2022).

Several works were conducted to evaluate the chemical profile of soil samples taken from the 2004 Indian Ocean tsunami-impacted regions in Thailand (Szczuciński *et al*. 2007, 2012) and Indonesia (Chaerun *et al*. 2009), as well as the 2011 Tohoku tsunami-impacted region in Japan (Chagué-Goff *et al*. 2015, 2017). Previous work for chemical analysis of soil samples taken from two worst-stricken areas by the 2004 Indian Ocean tsunami – namely, Banda Aceh City and Aceh Besar Regency – revealed that even after 1.5 years (2006) and 3 years (2008), the tsunami occurred where, in fact, there had been a natural bioremediation process after a long time. However, the concentration of salt and heavy metals was still very high (Chaerun *et al.* 2009). In addition, a recent qualitative study also revealed that the soil sample collected from the inundated area in Banda Aceh City 10 years after the 2004 Indian Ocean tsunami still contains salt elements – namely, Ca, Mg, Na, and K – at elevated levels. These elements mostly appeared in the soil samples from the impacted area compared with the unimpacted area (Mitaphonna *et al.* 2021).

The composition and concentration of chemical elements in tsunami deposits can change over time, making it difficult to distinguish them from terrestrial and marine incursions. Moreover, a single element in stratigraphy sometimes does not always yield a definite conclusion. Previous research used the elemental concentration ratio to investigate tsunami inundation (Watanabe *et al.* 2020; Moreira *et al.* 2017). A geochemical proxy of Na/Ti was reported as a good signature for investigating modern and past tsunamis in Tohoku, Japan (Watanabe *et al*. 2020). The high concentration ratio of Ca/Ti has been applied to the marine marker of the 1960 Chile tsunami in Laverick Bay, New Zealand (Donnelly *et al.* 2017). Moreover, the 1755 Lisbon tsunami deposit layer on the coast of Portugal is characterized by an increased ratio of Si/Al and Ca/Ti concentrations compared to the mud layer above and below it (Moreira *et al*. 2017).

Many analytical techniques have been employed to specify and quantify the chemical elements in samples – including atomic fluorescence spectrometry (Ferreira *et al*. 2019), flame atomic absorption spectrometry (Ayyildiz *et al.* 2021), inductively coupled plasma-mass spectroscopy (Xu *et al*. 2022), and X-ray fluorescence (XRF) (Ravansari *et al*. 2020; Al-Musawi and Kaczmarek 2020). Recently, an emerging analytical tool – namely, laser-induced breakdown spectroscopy (LIBS) adopting a fourth harmonic neodymium aluminum garnet laser – has been employed for studying the geochemistry of soil samples taken from several tsunami-stricken regions in Aceh Province (Idris *et al*. 2022). Since the intensity of the spectral emission lines is proportional to the concentration of analyte in the sample, the intensity ratio of the emission lines due to elements in the soil samples has been evaluated. It was found that the intensity ratio of a series of elements – namely, the ratios of Sr/Ba, Al/Ti, and Si/Ti – shows distinctly different values among the ratios of various other element pairs in the tsunami-affected soil sample compared to that in the non-tsunami-affected soil. This suggested that the ratio could be used for identifying the tsunami inundation area in Aceh. LIBS is a superior analytical tool with an excellent capability for conducting qualitative and quantitative chemical analysis of various samples, including difficult samples

such as soil, *etc.* However, LIBS has no capability for identifying the morphology of the sample. Moreover, an accurate quantitative analysis using LIBS requires standard samples with a matrix similar to the unknown sample. In the case of tsunami-impacted soil samples, not only is the chemical profile markedly altered, but the morphology of the sample is also strongly affected. Thus, the evaluation is not only focused on the chemical content but also on the morphology of the samples.

Scanning electron microscopy–energy dispersive X-ray (SEM-EDX) is a microscopic technique for observing the sample morphology. In addition to being able to image the topography of the sample surface in an adequate area using SEM, the composition and percentage concentrations of constituents can also be determined using EDX without the need for standard samples. Thus, chemical investigation using SEM EDX provides also an understanding of the morphology and topography of the surface of the investigated samples. SEM mainly utilizes electrons to create an image by scanning the sample surface with a focused electron beam through various magnifications depending on the need. The interaction of electrons with the atoms in the specimen generates various signals that contain details about the topography and composition of the surface. EDX is commonly employed for the chemical elemental investigation of materials. Identification of chemical constituents using EDX is based on the characteristics of X-rays emitted by secondary electrons. When an electron interacts with a material, it is scattered by other electrons surrounding the atomic nucleus. Electrons scattered in their path will be reflected out of the system, resulting in an electron vacancy that will be filled by electrons from the outer shell. The outermost electron has a higher energy so that when it moves to a lower energy orbit, it will release energy in the form of photons (X-rays). The energy spectrum of the emitted X-rays has a specific energy that depends on the atomic number of an element. By knowing the energy of the X-rays emitted, it is possible to know the atomic number and also the relative content of each material (Kayode *et al*. 2018).

SEM-EDX has been used in many applications such as materials science, life sciences, forensic science, corrosion science, metallurgical and electronic materials failure analysis, nanodevices, semiconductor design, polymer science, catalysis, desalination, oil and gas, mining, rock mineralogy, geoscience, and also in a case of occupational death (Aquila *et al.* 2020; Kikkawa *et al.* 2019; Kayode *et al.* 2018). Recently, SEM-EDX has been utilized for evaluating the chemical elements in sediments bought by the 2011 Tohoku tsunami in Sendai Bay, Japan (Abe *et al*. 2022). Therefore, in this study, SEM-EDX was adopted for carrying out the geochemical analysis of soil samples affected by the 2004 Indian Ocean tsunami in Aceh. This research is expected to know the capability of SEM EDX for morphologic and geochemistry studies of the soil samples impacted by the 2004 Indian Ocean tsunami. This work might contribute to adding the historical record of tsunamis and provide a scientific comparison of the past tsunamis in the region. This kind of study also could provide knowledge to enhance the bioremediation and revitalization processes of affected agricultural land in all tsunami-stricken areas in Aceh (Chaerun *et al*. 2009). In this study, XRF was employed to verify the elemental composition of the sample and as a benchmark of reliability of SEM-EDX for analysis of the tsunami-impacted soil.

MATERIALS AND METHODS

Geological Setting of Aceh Province

The high vulnerability to seismic activity in many areas in Indonesia, especially in Aceh Province, is due to the fact that it is located above the Holocene fluvial basin sediment and is surrounded by many fault segments. Aceh Province, located at the tip of Sumatra Island, is currently affected by a subduction system, where the Indian Ocean Plate is slowly crashing under the Eurasian Plate. The main structural elements and the surrounding area are determined by the Sumatra subduction system – namely, the forearc area consisting of the subduction trough zone, Bukit Barisan Mountain, and Sumatran fault, which is the product of volcanic activity bounded to the subduction zone. The back-arc area is occupied by tertiary sedimentary basins formed by rifting and Palaeogene subsidence. Moreover, Aceh Province is also flanked by four main segments – namely, the Tripa segment (180 km long), the Aceh segment (200 km long), the Seulimeum segment (120 km long), and the Batee segment (150 km long) (Sieh *et al*. 2015; Kelsey *et al.* 2015).

Tsunamigenic sediments are generally deposited in lowland coastal environments from silt to boulders mixed with organic debris from land and sea. Offshore underwater tsunamigenic sediments show intercalated coarser particles, mixed with shells and fragments of terrestrial plants, embedded within mud layers typical of deep-sea environments. Texture and structural characteristics of the sediments can also reveal depositional conditions underwater. Similar to the underwater beach environment, the sedimentation of tsunamigenic materials is observed by the deposition of suspensions in the water column resulting in a fining-upward sequence.

Sample Collection and Processing

The tsunami soil samples were taken in June 2014, nearly 10 years after the 2004 Indian Ocean tsunami. Figure 1 shows the map of sampling locations of soil samples struck by the 2004 Indian Ocean tsunami in the Aceh region – namely, Banda Aceh City, Aceh Besar, and Aceh Barat Regencies. As shown in Figure 1, these areas are directly facing the Indian Ocean. Thus, they are the most severely affected by tsunami disasters, where tsunami water enters the land even up to 6–7 km in certain areas, thus leaving seawater and mud from the coast with a thickness of 0.2–1 m on the land in their path. The soils were sampled from preserved areas, having a concave morphology, protected from coastal waves, and about 2–3 km away from the coastal line.

Banda Aceh City, situated in the western section of Sumatra, is the metropolis of Aceh Province. It was extremely destroyed and inundated by the tsunami seawater so far as 6–7 km in the hinterland. The coastal communities in Aceh Besar Regency were also shattered severely by the tsunami disaster. Aceh Barat Regency is the closest region to the epicenter of the Sumatra-Andaman earthquake that occurred on 26 Dec 2004, triggering the giant 2004 Indian Ocean tsunami. Soil samples from Aceh Besar Regency were collected in paddy fields that were submerged in seawater during a tsunami. Moreover, soil samples in Banda Aceh City were collected from vacant areas near residential homes, whereas those in Aceh Barat Regency were collected from vacant land close to the coast.

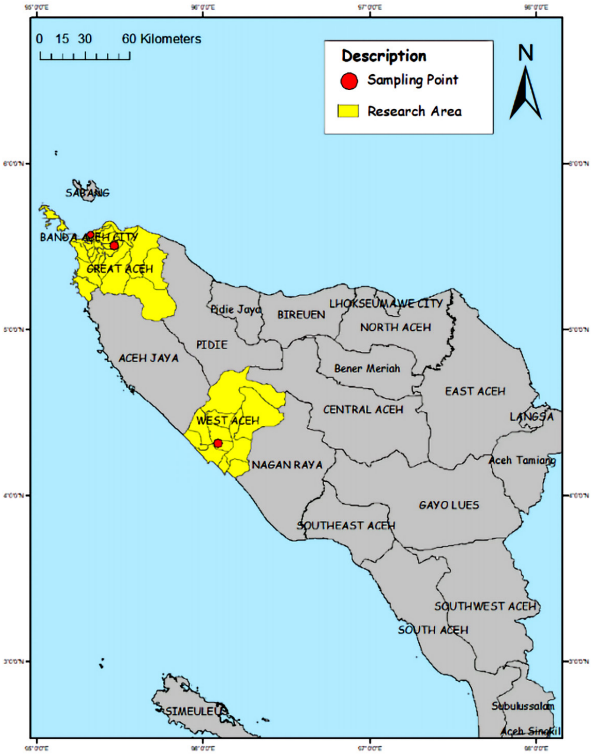

Figure 1. Map of sampling locations from soil sampled from affected regions by the 2004 Indian Ocean tsunami in Aceh Province, Indonesia. The sampling area is highlighted in yellow, and the sampling point is highlighted in red.

In previous work (Idris *et al*. 2022), we collected soil samples from several points in three regions – namely, three points in Banda Aceh City (Banda Aceh 4, Banda Aceh 5, and Banda Aceh 8), four points in Aceh Besar Regency (Aceh Besar 1, Aceh Besar 2, Aceh Besar 3, and Aceh Besar 9), and four points in Aceh Barat Regency (Kp.Cot 4, Kp.Cot 5, Kp.Cot 6, and Kp.Cot 7). Since the elemental concentration ratio was used to investigate tsunami inundation (Watanabe *et al*. 2020; Moreira *et al*. 2017), in the previous work (Idris *et al*. 2022), the chemical signature of tsunami event was evaluated using the intensity ratio of LIBS emission spectral lines with a consideration that intensity of the spectral emission lines detected LIBS is proportional to the concentration of the analyte in the sample. Thus, the emission intensity ratio was evaluated for several elements. It was found that the samples affected by the 2004 Indian Ocean tsunami show a distinctively high concentration ratio of the spectral line intensity of LIBS emission. The remarkably high ratios were found for Si/Ti, Al/Ti, Sr/Ba, Ca/Ti, and Fe/Ti. For example, the emission intensity ratio of Si/Ti was found to be 20 for the soil sample taken from Banda Aceh 5, the highest among the samples taken from the Banda Aceh region. The highest ratio of Si/Ti was found in Aceh Besar 1 (at 50), the highest among the samples taken from the Aceh Besar region. While among the samples taken from the Aceh Barat region, the highest ratio of Si/Ti was found in Kp. Cot 6 (at 32). The highest emission intensity ratio of Al/Ti was also found in Banda Aceh 5 sample (at 11) among the sample collected from the Banda Aceh region, in Aceh Besar 1 sample (at 26) among the samples collected from the Aceh Besar region and in Kp. Cot 6 sample (at 26) among the sample collected from the Aceh Barat region. A similar tendency was also observed for other elements – including Ca/Ti, Fe/Ti, and Sr/Ba – whereas for the other samples, the intensity ratio is lower. Therefore, we have selected three soil samples – namely, Banda Aceh 5 from Banda Aceh City, Aceh Besar 1 from Aceh Besar Regency, and Kp. Cot 6 from Aceh Barat Regency – to be investigated in this work for morphologic observation and further chemical analysis using SEM-EDX. In this present work, we write the samples as the Banda Aceh sample, the Aceh Besar sample, and the Aceh Barat sample, respectively.

The soil sampling was carried out using the drilling method by inserting a hand auger into the ground at each

predetermined point. Furthermore, the hand auger is rotated to the right and slowly pulled above the ground so that the sediment in the hand auger does not fall. After the soil sample is obtained, the samples are visually evaluated to observe sediment grain sizes, colors, and organic materials contained in them. The investigation of soil samples was carried out based on the characteristics of Troels-Smith (1955).

Referring to research conducted by previous researchers in Aceh Province, the 2004 Indian Ocean tsunami deposit layer was found with varying depths. Previous work found that the 2004 Indian Ocean tsunami deposit layer in a well-preserved coastal cave located in the western part of Aceh. The thickness of the 2004 Indian Ocean tsunami deposit layer is 20–40 cm (Rubin *et al*. 2017). Another study found the 2004 Indian Ocean tsunami deposit layer in an open area of Lhoknga coast, placed in Aceh Besar Regency, Aceh Province. The thickness of the tsunami layer is 5–20 cm (Moore *et al*. 2008). Recent work using a ground penetrating radar technique identified the 2004 Indian Ocean tsunami deposit in Meulaboh City, the capital city of Aceh Barat Regency. The tsunami layer was at a depth of 10–300 cm (Syukri *et al*. 2022). Considering the thickness of the 2004 Indian Ocean tsunami deposit found in those studies and the soil sampling made in an open area and also the sampling time of soil conducted more than 10 years after the tsunami, we can safely remove 5 cm of the surface layer to avoid any possible anthropogenic pollution. Additionally, based on our visual analysis, at a depth of 0-5 cm, we found dark brown peat soil and organic matter such as plants and roots. This layer also has uniform grain size from the bottom to the top of the layer. It is well-sorted and does not show the characteristics of the sediment carried by the tsunami.

After the identification, the soil samples were dried under sunlight for 1–2 days to remove moisture content. To obtain homogeneous particles, the dried samples were ground using a mortar and pestle. Furthermore, each sample was divided into three parts – namely, for analysis using SEM-EDX and XRF and also for analysis of grain size distribution. To investigate the grain size distribution of these materials as well as sand, silt, and clay, we utilized a particle sieve of 200 (0.074 mm). In our case, the identification of soil layers affected by the tsunami can be accomplished using a 200 mesh, as described below.

Weighing the dried samples was carried out in two repetitions – namely, for the particle sizes of ≥ 0.074 and ≤ 0.074 mm, respectively. To get the percentage of each sample measuring ≥ 0.074 and ≤0.074 mm, the weight of each sediment sample is divided by the total weight. Based on the result, the percentage of sand is obtained from the particle with a grain size of ≥ 0.074 mm, whereas the combined percentage of silt and clay is calculated from the particle with a grain size of ≤ 0.074 mm. The remaining soils with a grain size of ≤ 0.074 mm were reprocessed to obtain the percentages of silt and clay. This separation was carried out based on the grain size of the Wentworth scale. The silt and the clay have particle sizes of 0.074 and 0.003 mm, respectively. Therefore, in order to get the percentage of silt and clay in a sample with a particle size of ≤ 0.074 mm, the calculation is made by dividing the value of the grain size of silt (0.074 mm) or clay (0.003 mm) by the total result of the sum of the grain sizes of silt and clay (0.077 mm). It is considered that each sediment with a grain size of ≤ 0.074 mm consist of 96% silt and 4% clay. After the sieving process was completed, the sample was classified into sand, silt, and clay sediments.

SEM-EDX Analysis

The chemical investigation of tsunami-impacted soil samples was performed using SEM-EDX (Thermo Fisher Scientific, FEI, Inspect-S50). Initially, those soil samples were dried in the oven for 3 d at 60 °C to omit the water in the samples. Furthermore, the soil samples were ground and sifted for gaining particle sizes of ≤ 1 mm. The 5.0 g of the sample was weighed and mixed with 25 mL of deionized water in a conical flask. Then, the soil samples were stirred in an ultrasonic bath for 7 h. SEM gathered with the Thermo EDX spectrophotometer and detector of cathode luminescence was utilized to investigate the chemical composition of the soil samples. The X-ray was focused on soil placed in a disc of polyethylene with the recording of soil surface photographs and analytical peaks.

RESULTS

SEM images of soil samples taken from three tsunami-stricken areas in Aceh Province are displayed in Figure 2. The SEM images show that 10 years after the tsunami struck Aceh Province, the surface topography of soils from the three regions is characterized by small particles and inhomogeneous particle sizes. Based on our visual analysis using the Troels-Smith classification, the subsoil affected by the tsunami in Aceh Province consists of very coarse sand at the bottom of the layer, which decreases gradually to fine-grained clay silt in the upper part of the layer. The weakening of the current during the backwash tsunami causes fine-sized particles such as clay and silt to be deposited on the surface of the layer. Furthermore, the soil surfaces became hardened in all-stricken regions compared to the unimpacted regions.

In addition, geochemical analysis in tsunami-impacted soil samples was further identified using EDX (Figure 3). Based on the obtained EDX peaks, it could be noticed that 10 years after being flooded by the 2004 Indian Ocean

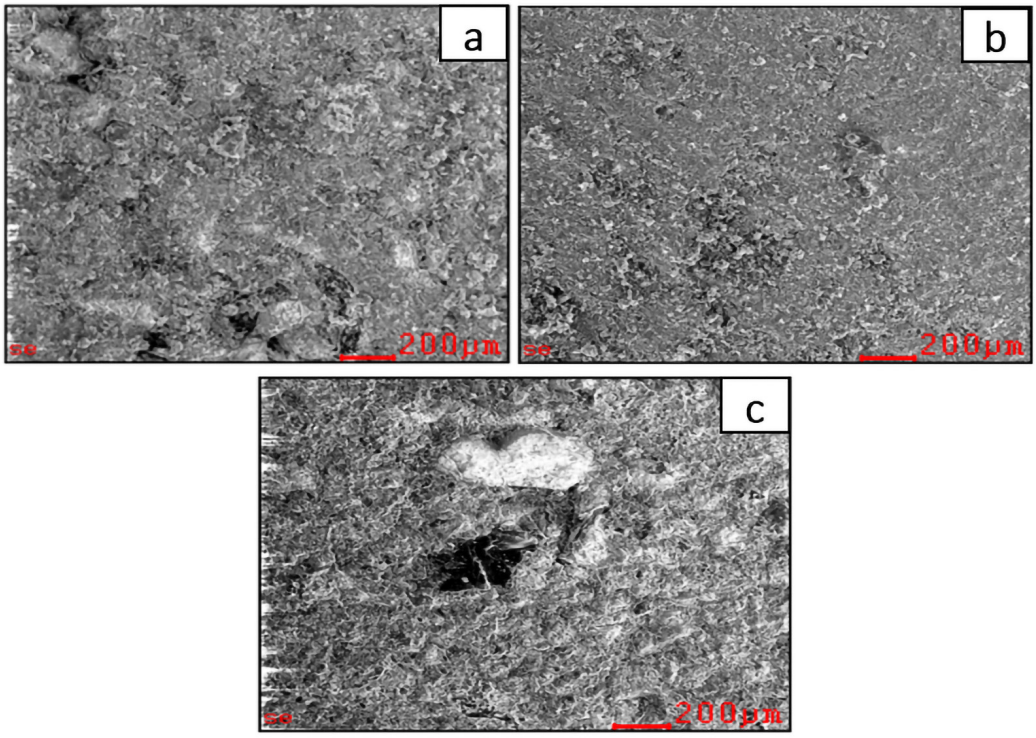

Figure 2. SEM images from soil sampled from affected regions in Aceh Province: [a] Aceh Besar Regency, [b] Banda Aceh City, and [c] Aceh Barat Regency.

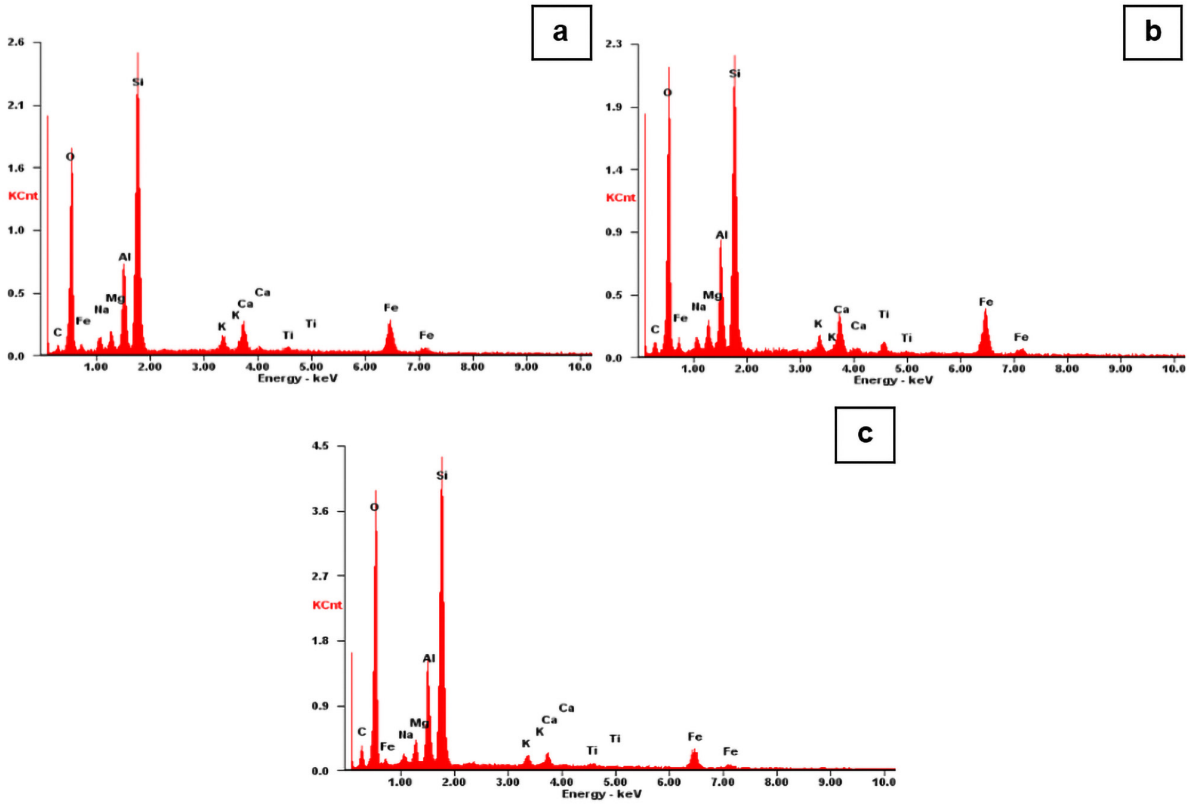

Figure 3. EDX peaks of soil sampled from regions affected by the 2004 Indian Ocean tsunami in Aceh Province: [a] Aceh Besar Regency, [b] Banda Aceh City, and [c] Aceh Barat Regency.

tsunami, those soils from the impacted regions were still polluted by organic elements O and C; salt elements Na, Mg, Ca, and K; and also metal elements Al, Ti, Fe, and Si. However, the EDX measurement applied in this experiment could not be used to identify heavy metal elements (Ba, Pb, Cu, Co, and Cd) and low-molecular-weight elements (H, He, Li, Be, B, and N). The elemental concentration of the tsunami-affected soil samples collected 10 years after the disaster is demonstrated in Table 1. Moreover, the level of chemical elements in farming land in Banda Aceh City 1.5 and 3 years after the 2004 Indian Ocean tsunami published earlier by Chaerun *et al*. (2009) is shown in Table 2. The measurement was detected by electrical conductivity.

Table 1. Concentration of chemical elements detected using EDX in soil sampled from regions affected by the 2004 Indian Ocean tsunami 10 years after the tsunami in Aceh Province, Indonesia.

Element	Concentration (%)		
	Aceh Besar	Banda Aceh	Aceh Barat
C	11.59 ± 0.34	16.42 ± 0.37	23.37 ± 0.22
O	54.86 ± 0.85	53.68 ± 0.42	52.57 ± 0.47
Na	1.85 ± 0.09	1.59 ± 0.09	1.01 ± 0.10
Mg	1.39 ± 0.02	1.78 ± 0.08	1.36 ± 0.04
Al	6.28 ± 0.22	5.67 ± 0.11	5.07 ± 0.16
Si	17.35 ± 0.37	13.41 ± 0.07	13.37 ± 0.08
K	0.88 ± 0.04	0.72 ± 0.05	0.49 ± 0.04
Ca	1.74 ± 0.12	1.83 ± 0.05	0.58 ± 0.04
Ti	0.32 ± 0.03	0.59 ± 0.03	0.19 ± 0.005
Fe	3.73 ± 0.43	4.38 ± 0.14	1.96 ± 0.01

The chemical composition of the soil samples from affected areas in Aceh Province included 50% of O, followed by C and Si. The maximum content of O was found in a soil sample from Aceh Besar Regency with a value of 54.86 ± 0.85%. Moreover, the concentration of C was twice as high in impacted soil from Aceh Barat Regency (23.37 ± 0.22%). The level of Na in the tsunami-impacted soil collected from Banda Aceh City 1.5 years after the tsunami was 0.12%, which further decreased 3 years after the event (Chaerun *et al*. 2009) (Table 2). Meanwhile, the levels of Na in the soil samples collected from Aceh Besar Regency and Banda Aceh City 10 years after the tsunami were 1.85 ± 0.09 and 1.59 ± 0.09%, respectively. As shown in Table 1, the maximum concentrations of Mg and Ca appeared in Banda Aceh City, whereas the level of K mostly occurred in Aceh Besar Regency. It was found that Al and Si dominantly

Table 3. XRF analysis of tsunami-impacted soil collected from several regions affected by the 2004 Indian Ocean tsunami in Aceh Province, Indonesia

Element	Concentration (%)		
	Aceh Besar	Banda Aceh	Aceh Barat
C	–	–	–
O	–	–	–
Na	–	–	–
Mg	–	–	–
Al	4.96	4.88	4.36
Si	15.56	14.27	12.28
K	0.91	0.84	0.80
Ca	1.83	1.91	1.60
Ti	0.70	0.60	0.56
Fe	3.77	4.42	3.58
Ni	0.20	0.17	0.21
Mn	0.43	0.22	0.88

appeared in the tsunami soil sample from Aceh Besar Regency. The content of Si was considerably lower (only 0.28% remaining) in the tsunami soil samples collected 3 years after the tsunami. Furthermore, the maximum contents of Fe and Ti were found in the soil sample from Banda Aceh City, and the lowest concentrations were in the Aceh Barat Regency sample.

For verification, the presence and the concentration of chemical elements were revealed by EDX measurement, and the soil samples were also analyzed using XRF. The results from the XRF measurement are shown in Table 3. In general, XRF measurement results are in accordance with the EDX results, confirming the presence of chemical markers of marine incursion on the tsunami-affected soils – namely, Ca, K, Al, Si, Fe, Ti, Mn, and Ni. The highest concentrations of Si, K, and Ti elements were found in the tsunami-affected soil samples collected from Aceh Besar Regency with values of 15.56, 0.91, and 0.70%, respectively.

Meanwhile, the concentrations of Ca and Fe were dominantly found in the tsunami-affected soil sample taken from Banda Aceh City compared to the other stricken areas with values of 1.91 and 4.42%. The tsunami-affected soil sample taken from the Aceh Barat Regency showed an increase in concentrations of Al, Ni, and Mn with values of 4.96, 0.21, and 0.88%, respectively. Although the XRF analysis used was able to detect salt

Table 2. Concentration of chemical elements in soil sampled from regions affected by the 2004 Indian Ocean tsunami 1.5 and 3 years after the tsunami in Aceh Province, Indonesia (Chaerun *et al*. 2017).

Chemical element	Location points			
	Tsunami-impacted soil (%)	Tsunami unimpacted soil (%)	Tsunami-impacted soil (%)	Tsunami unimpacted soil (%)
	May 2006	May 2006	June 2008	June 2008
C	0.00031	0.00025	0.00012	0.00012
O	–	–	–	–
Na	0.12	0.01	0.10	0.02
Mg	0.72	0.35	0.62	0.35
Al	5.69	5.08	4.42	3.59
Si	NM	NM	0.28	0.24
K	0.21	0.11	0.29	0.24
Ca	1.35	0.46	1.11	0.44
Ti	–	–	–	–
Fe	3.86	3.24	4.11	3.58

elements and heavy metal elements (Ni and Mn), however, XRF could not detect other salt elements (Na and Mg) and organic elements that have lower molecular weights (C and O).

Even though many references examined the concentration ratios of Na/Ti, Ca/Ti, Si/Al (Watanabe *et al*. 2020), Fe/Ti, Sr/Ba (Wang *et al*. 2021), Sr/Ti, and Sr/Rb (Watanabe *et al*. 2022) as geochemical markers for tsunami inundation, we also evaluated the concentration ratio of other chemical elements available in EDX measurement results – including Na/Ti, Ca/Mg, Ca/Ti, Ca/K, Ca/Fe, Mg/Ti, Mg/K, Fe/Ti, Fe/K, Si/K, Si/Al, and Si/Ti – as potential candidates for chemical signatures of tsunami influences for identifying the soil impacted by the 2004 Indian Ocean tsunami in Aceh Province, Indonesia. The elemental concentration ratios deduced from EDX measurement results are displayed in Table 4.

It can be seen in Table 4 that the highest Na/Ti concentration ratio was found in a sample collected from Aceh Besar Regency with a value of 6, and the lowest concentration ratio was in a sample from Banda Aceh City (at 3). The concentration ratios of Na/K and Na/Mg obtained in this study show the same value in all tsunami-affected areas (respectively at 2 and 1). The concentration ratio of Ca/Fe in this research work has a low value, 0.5, in all affected regions. The high ratio of Ca^{2+}/K^{+} dominantly appeared in a sample from Banda Aceh City with a value of 3. The maximum concentration ratio of K/Ti occurred in tsunami-affected soil in Aceh Besar and Aceh Barat Regencies with

Table 4. Concentration ratio of several elements detected using EDX in soil sampled from regions affected by the 2004 Indian Ocean tsunami in Aceh Province, Indonesia.

Elemental ratio	Locations of soil sampling		
	Aceh Besar	Banda Aceh	Aceh Barat
Na/Ti	6	3	5
Na/K	2	2	2
Na/Mg	1	1	1
Ca/Mg	1	1	1
Ca/Ti	5	3	3
Ca/Fe	1	0.5	0.3
Ca/K	2	3	1
Mg/Ti	4	3	7
Mg/K	1	2	3
K/Ti	3	1	3
Fe/Ti	11	7	10
Fe/Al	1	1	0.5
Si/K	20	19	27
Si/Al	3	2	3
Si/Ti	54	23	70

a value of 3. A similar tendency is also observed for several elemental ratios – namely Ca/Ti, Ca/Fe, Mg/Ti, Mg/K, Fe/Al, and Si/Al. Thus, these concentration ratios are not suitable to be used as chemical signatures for identifying the soil samples struck by the 2004 Indian Ocean tsunami in Aceh Province. The highest concentration ratio of Fe/Ti appeared in a soil sample from Aceh Besar Regency with a value of 11. Furthermore, the concentration ratios of Si/Ti and Si/K in a soil sample from Aceh Barat Regency obtained using EDX have a value of 70 and 27, respectively. These concentration ratios – namely Si/Ti, Si/K, and Fe/Ti – have distinctively different values in all tsunami-affected areas in Aceh Province; therefore, it can be suggested as a geochemical marker of the 2004 Indian Ocean tsunami inundation.

Moreover, we also made a similar evaluation of the concentration ratios for several chemical elements using XRF measurement results as displayed in Table 5. It is found the same values for the elemental concentration ratios of Ca/Ti, Ca/Fe, Ca/K, K/Ti, and Si/Al in all soil samples affected by the Indian Ocean tsunami with values of 3, 0.5, 2, 1, and 3, respectively. Thus, these ratios are not suitable for use as geochemical markers of tsunami-affected soils. Furthermore, the concentration ratio of Fe/Ti was dominantly detected in the tsunami-affected sample soil collected from the Aceh Besar Regency with a value of 7. The high concentration ratio of Si/K appeared

Table 5. The concentration ratio of chemical elements detected using XRF in soil sampled from regions affected by the 2004 Indian Ocean tsunami in Aceh province, Indonesia.

Elemental ratio	Locations of soil sampling		
	Aceh Besar	Banda Aceh	Aceh Barat
Ca/Ti	3	3	3
Ca/Fe	0.5	0.5	0.5
Ca/K	2	2	2
Ni/Ti	0.5	0.3	0.3
Mn/Ti	1	0.3	1
K/Ti	1	1	1
Fe/Ti	5	7	6
Fe/Al	0.5	1	0.5
Si/K	17	17	16
Si/Al	3	3	3
Si/Ti	22	24	23

with values of 17. Meanwhile, the concentration ratio of Si/Ti is the highest among all the chemical element concentration ratios evaluated, with a value of 24 in a soil sample from Banda Aceh City. Thus, it can be concluded that the elemental concentration ratios of Fe/Ti, Si/K, and Si/Ti are suitable geochemical markers to investigate the 2004 Indian Ocean tsunami inundation in Aceh Province. Although the concentration ratio values are not exactly the same, the pattern of the concentration ratio of various elements obtained from XRF measurement results closely matches the concentration ratio pattern obtained from EDX measurement results above (Table 4).

DISCUSSIONS

The results show that the soil samples taken from Aceh Besar and Aceh Barat Regencies tend to have coarse to fine grain-sizes sand may be related to their proximity to the coast area, which caused tsunami waves to transport large particles. Meanwhile, the collected sample from Banda Aceh City had a smooth texture of sand. The difference in tsunami soil texture in Aceh Besar Regency and Banda Aceh City is supposed due to the sampling location in Aceh Besar Regency being close to the coast, which was about 1 km from the coastal line, so sea waves were able to carry coarse sand materials. Tsunami deposits are thinner and have a finer grain size near the inundation boundary because of the weakened capacity of the tsunami waves to carry large amounts of coarse sand particles (Shinozaki *et al*. 2021). In general, tsunami deposit has a thickness and grain size that decreases inland, although it also depends on the local topography and elevation of the land (Yoshii *et al*. 2018; Widiyanto *et al*. 2020). The lower slope of the land, the concentration of tsunami deposits increases and supplies large amounts of coarse sand (Yoshii *et al*. 2018). The results obtained in this work are in accordance with the finding in the coastal region of Thailand (Szczuciński *et al*. 2007). The tsunami soil samples in Patong City, Thailand, had a relatively coarser size. This result can be ascribed to its proximity to the groove river and the natural path of tsunami wave propagation. Thus, this location had a stronger speed and stronger carrying capacity than the other locations. The 2011 Tohoku tsunami deposits found in Suijin-numa – a small lake in the Sendai plains, Japan – were observed as very fine to medium-sized sand under a layer of black silt (Shinozaki *et al*. 2015) The tsunami deposition on the coast of the Sunda Strait, Indonesia, is marked by fine sand to gravel, which is dominated by medium and well-separated coarse sand (Widiyanto *et al*. 2020). Moreover, the tsunami deposits at Ladang Tuha district, Aceh Selatan Regency, Indonesia are characterized by a coarse sand layer between peat layers with firm contact and brownish-grey in color (Putra *et al.* 2013).

The soil surface became hardened in all tsunami-affected areas in Aceh Province could be considered as a result of the sedimentation of dissolved chemical elements during the tsunami flooding. Seawater contains high concentrations of salt components (> 3.5%), mainly in the form of NaCl, a combination of cations bases including potassium (K), magnesium (Mg), calcium (Ca), sulfate ($SO_4 2^-$), bicarbonate (CO_3^{2-}), and chlorine (Cl^-). The tsunami incident in Aceh Province not only inundated the land with seawater but also carried mud and sand from the seafloor and then precipitated it on agricultural land, basins, and other places with a thickness of around < 5 to 20 cm. In addition to containing salts that have the potential to increase soil salinity, tsunami mud also contained high organic materials, carbon (C), and cations (Na, Ca, Mg, K). In mud, salt elements can infiltrate the soil, which has the potential to elevate the salinity of roots, damage soil structures, and contaminate groundwater. The higher content of salt in the soil, the more easily the soil is dispersed. Dispersed soil particles move to clog the soil pores, causing the soil to compact and decreasing the oxygen supply for root growth and soil microbes. This condition also inhibits infiltration, allowing very little water to enter the soil and most water to stagnate on the surface, thereby affecting the surface hardening of the soil. The results of this experiment agree with those obtained in previous studies carried out on agricultural land in northeast Japan, which also revealed that the soil surface became salty and hardened in the tsunami-affected area because of the deposition of high salt elements and sediment materials (Roy *et al*. 2014). Previous literature

also reported that the soil surface in the agricultural land in Banda Aceh City became hardened because of the high concentrations of sodium (Agus *et al*. 2012).

The higher the salt content in the soil, the easier it is to dissolve. Dispersed soil particles move to clog the pores of the soil, causing the soil to become compact and reducing the supply of oxygen for the growth of roots and soil microbes. This condition also inhibits infiltration, so that very little water enters the soil and most of the water stagnates on the surface, thereby affecting the hardening of the soil surface. The results of this experiment are in line with previous studies conducted on agricultural land in northeastern Japan, which also revealed that surface soils become saline and harden in the tsunami-affected areas due to the deposition of high salt elements and sedimentary materials (Roy *et al*. 2014). Previous literature also reported that the soil surface on agricultural land in Banda Aceh City became hardened due to high sodium concentrations (Agus *et al.* 2012). In addition, the paddy fields in Banda Aceh City experienced crop failure due to the deposition of mud from the seafloor – which is rich in salts, organic elements, and heavy metals (Sembiring *et al*. 2008).

The soil properties can also be described in terms of color because of the influence of various factors. Differences in the color of soil surface are generally caused by organic matter content. The higher the organic and metal contents in the soil, the darker the color of the soil. The color of the affected soil in Aceh Province changed from brown to grayish-black, possibly because of the oxidation-reduction reaction of metallic elements such as Fe in the soil. In poor drainage areas that were always flooded, the soil was gray because it contained reduced Fe (Fe^{2+}). However, in well-drained soil, Fe was present in an oxidation state (Fe^{3+}) such as a red compound of Fe_2O_3 (hematite) or $Fe_2O_3 \cdot 3H_2O$ (limonite). If the soil was sometimes dry and wet, in addition to being grey (reduced area), red or yellow rusty spots were also obtained because of the entry of air so that iron oxidation reactions occurred in the soil.

The purpose of geochemical analysis with EDX in this study is to investigate the different environments in terrestrial and marine deposits and also to identify the geochemical signatures of tsunami-affected soil. In Figure 3, EDX analysis revealed that the tsunami-affected soil consists of Si, O, Al, Mg, C, Ca, K, Na, Ti, and Fe in various percent concentrations. This result looks almost similar to the EDX result conducted by Philip and Singh (2020), who showed that the Canterbury (United Kingdom) soil was formed by Si, Al, O, Mg, Ca, Ti, Fe, and K. The presence of Na concentration in EDX analysis confirmed that elemental salt enrichment had occurred in the tsunami-affected areas compared to normal soils. Sharma *et al*. (2016) also explained that the concentration of Na and the other eight elements elevated in the area of anthropogenic activity. By using EDX analysis in SEM-EDX, it has been found that Na is an element capable of distinguishing between tsunami-affected and non-tsunami areas. As discussed previously, the results showed a similar tendency to that found in soil samples taken from several affected areas in Aceh Province just after the 2004 Indian Ocean tsunami hit. Several previous studies also found that the salt and heavy metal contents in the tsunami-affected soils are much higher compared to the soil samples from unimpacted areas (Idris *et al.* 2017, 2018). Paddy fields in Aceh Province contain high levels of salt (Na^+, Mg^{2+}, Ca^{2+}, and K^+) because some of the tsunami sediments and seawater are trapped in long-term conditions (Agus *et al.* 2012).

The increase in C and O in this study area 10 years after the tsunami event could be assumed due to the high concentrations of calcite ($CaCO_3$) and dolomite $CaMg(CO_3)_2$ minerals carried by the tsunami seawater. As discussed previously, C content in tsunami-impacted soil in Lampisang district, Aceh Besar Regency, Indonesia after 1.5 years of tsunami is only 0.00031%. This result can be attributed to the mineral composition in the region, where quartz (SiO_2) and plagioclase ($NaAlSi_3O_8$-$CaAl_2Si_2O_8$) were the major minerals in tsunami sediments (Chaerun *et al*. 2009).

The geochemical signatures of tsunami deposits in coastal New Zealand are marked by the presence of Ca and Sr (Chagué-Goff *et al*. 2020). The increase in concentrations of Na, Cl, S, and Sr in the tsunami deposits indicates saltwater signals (Mitaphonna *et al.* 2021; Idris *et al*. 2022). Na contents could be supplied from sea salt and feldspar minerals from marine and coastal floors during tsunami waves (Watanabe *et al.* 2020). Three years after the 2004 Indian Ocean tsunami, Na elements possibly leached and disappeared from the soil layer. As a result, the concentration of Na decreased in tsunami-impacted soil in Banda Aceh City. The leaching of Na depends on topography, regional characteristics, and environmental condition after deposition. However, the increase in Na concentration after 10 years of the 2004 Indian Ocean tsunami was assumed due to the sampling region being located in a basin area and muddy soil, which caused Na accumulation and preservation in its area. The concentration of Na increased in the muddy soil up to 1.6%, whereas the concentration of Na was much lower in the sandy layer (0.04–0.1%) (Chagué-Goff *et al*. 2011). Na element might have been leached from precipitating the authentic tsunami, and its chemical composition may gradually turn into a rocky material (Watanabe *et al*. 2020). Some minerals containing Na such as feldspar and mica can be well maintained in tsunami sediment even more than 2000 years, although a small proportion

of Na has been leached from previous tsunami sediments (Watanabe *et al*. 2020). Tsunami waves brought marine sediments containing salt elements (Mg, Ca, and K) produced from shelled animals – namely, ostracods, foraminifera, or gastropods – to inundated land. The concentration of salt elements in soil samples taken from the Aceh Besar Regency and Banda Aceh City were higher than those from the Aceh Barat Regency. This phenomenon may be attributed to the fact that the soil samples collected from these areas were muddy, had a fine texture, and were rich in organic matter. Thus, the soil samples from these areas can hold higher salt content than the soil samples collected from the Aceh Barat Regency, which has porous sand. Almost all salt contents involving Na, Mg, Ca, K, and PO_4^{3-} had low concentrations in tsunami sand deposits because their porosity gradually increased under a tsunami sand layer (Shinozaki *et al*. 2016). This condition indicates that the chemical markers of seawater cannot be preserved in the sand because of the leaching process by high rainfall. Soil has a finer texture and is rich in organic matter, which can hold more water content so that the chemical components of seawater become concentrated in the soil. In addition, prominent salts – namely, Cl, S, and other water-soluble ions – were not detected in the tsunami-affected soil samples. The absence of this component is probably due to differences in the permeability of mud and sand.

Water-soluble ions present in sandy tsunami deposits tend to be leached by rainwater seepage, and seasonal changes in groundwater (Shinozaki *et al*. 2016). A low concentration of water-soluble ions (Ca^{2+}, Cl^-, Na^+, K^+, Mg^{2+}, Rb^+, S^{2-}, and Sr^{2+}) is observed in sandy tsunami deposits, whereas the high concentrations were found in the soil located below tsunami deposit layers (Spiske *et al*. 2020). In addition, at other sites, high concentrations of soluble salts were found in sandy tsunami deposits with a thickness of 1 cm due to the long-term inundation of seawater after the tsunami event (Chagué-Goff *et al*. 2012b). This event demonstrated that the salt component is generally soluble through a thick layer of sand. Si element can be applied as a chemical marker for sandy deposits carried by tsunami waves dominated by mineral assemblage silicates (Yu *et al*. 2016). Fe and Ti are commonly used as chemical proxies for terrestrial sources. However, these two elements are also the main constituents of heavy minerals as markers of high-energy tsunami waves.

All of the chemical element concentrations, including organic and salt elements in affected soils sampled from Aceh Province, appeared higher after 10 years of the tsunami compared with the research conducted 1.5 and 3 years after the tsunami. The concentrations of salt in soil affected by the 2004 Indian Ocean tsunami in Thailand decreased after the rainy season. In addition, the high concentration of water-soluble ions from the 2011 Tohoku tsunami was observed several months after the tsunami, although the higher concentrations no longer appeared several years later (Shinozaki *et al*. 2016). This result might be due to the different points of collecting samples, the coastal regions versus the agricultural land, and the time of collecting samples, which could be affected by environmental parameters such as rainfall and sunlight. Moreover, external factors (pollution and process of geology) possibly affect the different distributions of chemical elements. Conversely, the 2004 Indian Ocean tsunami event not only negatively affected the inundated land – such as an increase in salt and heavy metal elements – but also exerted positive impacts. The increasing micronutrients and macronutrients in organic carbon help decrease salt and heavy metal pollution in the soil.

The chemical composition after deposition could significantly change with time, making the deposits from marine and terrestrial markers difficult to distinguish. Moreover, the distribution of a single element in stratigraphy sometimes may not provide a clear conclusion. As discussed previously, Ca, Mg, K, and Sr sometimes can be used as markers of shell material or seawater signals because there is higher carbonate content in seawater than in freshwater and in the terrestrial environment; therefore, they can be applied to confirm tsunami flooding. However, not all shell materials come from the sea. The main source of Ca and Sr in Mangaia, Cook Islands after multiple tsunamis was from freshwater gastropods covering marine markers (Chagué-Goff *et al.* 2016a). Furthermore, the historical tsunami deposits in the eastern Mediterranean over the past 700 years detected in the sediments of Oludeniz Lagoon, southwest Turkey were dominated by K, Ti, Fe, and Zn but reduced in Ca. This phenomenon is assumed due to a large amount of terrigenous sediments enter into the lagoon (Avşar 2019). Therefore, the previous research used the elemental concentration ratios to discriminate the chemical signatures of tsunami deposits from their natural environment. Several chemical elemental concentration ratios – including Na/Ti (Watanabe *et al*. 2020), Ca/Ti (Willershäuser *et al*. 2015), Fe/Ti (Watanabe *et al.* 2020), and Sr/Ba (Donnelly *et al*. 2017) – have been widely applied to investigate the geochemical signature of tsunami events.

The soil samples collected from Sendai plain, Tohoku, Japan after the 2011 Tohoku tsunami showed a sharp increase in the concentration ratio of Na/Ti to 23.4, which was higher than that of unimpacted areas (at 10.3) (Watanabe *et al*. 2020). They also found that the geochemical proof of Na/Ti can be maintained for a long period so that such geochemical evidence may be applied as a signature for tsunami deposits in other locations.

The multiple prehistoric tsunami deposits in Lefkada lagoon, Greece are illustrated by a high concentration of $CaCO_3$ (> 45%) and increased Na/K and Ca/K ratios (May *et al*. 2012). Previous literature revealed that a high concentration ratio of Ca/Ti was a good signature for identifying prehistoric tsunamis (Willershäuser *et al*. 2015). In addition, Fe and Ti are terrigenous sediments that have low concentrations in seawater so they can be used as indicators of tsunami deposits. Fe is obtained from weathering of limestone and hematite minerals from rocks, whereas Ti is abundant in rutile and ilmenite mineral deposits. A high concentration ratio of Ca^{2+}/Mg^{2+} is a good indicator for investigating the effects of multiple prehistoric tsunamis in the Sound of Lefkada, Yunani, whereas the limnic and the terrestrial markers have a low concentration ratio of Ca^{2+}/Mg^{2+} (Vött *et al*. 2009). Tsunami deposits contain a high concentration ratio of Ca^{2+}/Mg^{2+} derived from calcium carbonate production by marine organisms *vs.* Mg-rich clay. The soil samples collected from tsunami-affected areas in Aceh Province after more than 10 years of the 2004 Indian Ocean tsunami showed a sharp increase in the ratio of Si/Ti, which is nine times higher than of soils not affected by the tsunami. Moreover, the intensity ratio of Sr/Ba increased to 5 times higher in tsunami-affected soils. In addition, the ratio of Al/Ti is found to increase in tsunami-affected soil (at 26), which is three times higher than in non-tsunami-affected soils (at 8) (Idris *et al*. 2022).

By contrast, the concentration ratios of Na/Ti, Na/K, Ca/Ti, Ca/Fe, Ca/Mg, Ca/K, Mg/K, Mg/Ti, K/Ti, and Si/Al in this work did not show clear differences in the case of tsunami-affected land (Table 4). Furthermore, the concentration ratios of Si/Ti and Si/K in a soil sample from Aceh Barat Regency and the concentration ratio of Fe/Ti from Aceh Besar Regency show the distinctively different values in all tsunami-affected soil samples collected from Aceh Province with values of 70, 27, and 11, respectively. This phenomenon is considered due to the closeness of the Aceh Barat region to the mega earthquake epicenter in the Indian Ocean so that it can carry large amounts of quartz sand into the region as compared to more distant areas. Furthermore, the distance from Aceh Barat Regency to the epicenter was estimated to be 49.4 km, whereas the distances from Banda Aceh City and Aceh Besar Regency were estimated to be 238 and 249.7 km, respectively. Our recent work of geochemistry study of soil sampled from tsunami impacted region in Aceh Province using the new superstar analytical tool – namely, LIBS – found that the concentration ratios of Si/Ti, Al/Ti, and Sr/Ba are a good geochemical proxy for the investigation of soil samples affected by the 2004 Indian Ocean tsunami in Aceh Province (Idris *et al*. 2022). The obtained results found in this present work using SEM EDX are just in good accordance with the previous research – namely, Si/Ti – is a suitable geochemical marker for the identification of soils affected by the 2004 Indian Ocean tsunami in Aceh province. In addition, SEM EDX measurement allowed morphologic evaluation of the investigated sample.

CONCLUSIONS

The chemical elements of soil sampled from the massive affected areas in Aceh Province 10 years after the 2004 Indian Ocean tsunami event were investigated using SEM-EDX. After more than 10 years of being inundated by the 2004 Indian Ocean tsunami, those soils from three tsunami-stricken regions in Aceh Province had inhomogeneous particle sizes consisting of coarse sand to fine silt. The soil surface in the tsunami-affected areas in Aceh Province experienced hardening due to the deposition of dissolved chemical elements during the tsunami floods. In addition, EDX measurement revealed that the tsunami-impacted soils still contain a high concentration of organic elements (O and C), salt elements (Na, Mg, Ca, and K), and metal components (Al, Ti, Fe, and Si). However, the SEM-EDX technique cannot be applied to identify the content of heavy metals (Ni Mn, Ba, Sr, Cd, Co, and Pb), and other chemical elements that have low molecular weights, such as H, He, Li, Be, B, and N in the soil samples investigated in this work. This condition is due to the possibility that heavy metals and other elements are not present in the tsunami-affected soil, or the concentration of these elements is low, so it is difficult to detect using EDX. Moreover, the chemical signatures have provided pieces of evidence of seawater inundation, animal shells, and heavy mineral pollution associated with tsunami waves. Our results suggest that a high concentration ratio of several elements, namely Si/Ti, Si/K, and Fe/Ti could be used as good geochemical signatures for predicting the massive Indian Ocean tsunami on 26 Dec 2004 in Aceh Province. This finding is just consistent with the result obtained by the LIBS instrument.

ACKNOWLEDGMENTS

The authors thank the financial support from the Directorate of Research and Community Service Directorate General of Research Strengthening and Development, Ministry of Research, Technology, and Higher Education, Government of Republic Indonesia, through Higher Education Primary Flagship Research Grants (Penelitian Dasar Unggulan Perguruan Tinggi, PDUPT) 2019 with a contract number: 215/SP2H/LT/DPRM on 08 Mar 2019, and Research of Master to Doctoral Education for Excellent Undergraduate (Program

Magister Doktor Sarjana Unggulan, PMDSU) 2020 with a contract number: 111/UN11.2.1/PT.01.03/DPRM/2020 on 04 Aug 2020.

STATEMENT ON CONFLICT OF INTEREST

The authors declare no conflict of interest.

REFERENCES

ABE H, KAWAMURA N, ISHIKAWA N, KOGISO T. 2022. Changes in elements and magnetic properties of Sendai Bay sediments caused by the 2011 Tohoku-oki tsunami. Island Arc. https://doi.org/10.1111/iar.12437

AGUS F, RACHMAN A, WAHYUNTO, RITUNG S, MCLEOD M, SLAVICH P. 2012. The dynamics of tsunami-affected soil properties in Aceh, Indonesia. Journal of Integrated Field Science 9: 11–20.

AL-MUSAWI M, KACZMAREK S. 2020. A new carbonate-specific quantification procedure for determining elemental concentrations from portable energy-dispersive X-ray fluorescence (PXRF) data. Appl Geochem 113: 104491.

AQUILA I, BOCA S, RICCI P, PEROZZIELLO G, CANDELORO P, FABRIZIO ED, MALARA N, GRATTERI S, SACCO MA, FRANCARDI M. 2020. The use of scanning electron microscopy and energy dispersive X-ray spectroscopy in a case of occupational death. The Medico Legal Journal 88(3): 002581721989108.

AVŞAR U. 2019. Sedimentary geochemical evidence of historical tsunamis in the Eastern Mediterranean from Ölüdeniz Lagoon, SW Turkey. J Paleolimnol 61: 373–385.

AYYILDIZ MF, YAZICI E, SAYLAN M, CHORMEY DS, BAKIRDERE S. 2021. Determination of copper in human blood serum by flame atomic absorption spectrometry after UV-assisted Fenton digestion using binary magnetite nanoparticles. Measurement 186: 110108.

CHAERUN SK, WHITMAN WB, WIRTH SJ, ELLERBROCK RH. 2009. Chemical and mineralogical characterization of agricultural soils inundated by the December 26, 2004 tsunami after intrinsic bioremediation in Banda Aceh, Sumatra Island, Indonesia. Journal of the American Society of Mining and Reclamation 40502 30: 3134.

CHAGUÉ-GOFF C, SCHNEIDER JL, GOFF JR, DOMINEY-HOWES D, STROTZ L. 2011. Expanding the proxy toolkit to help identify past events – lessons from the 2004 Indian Ocean tsunami and the 2009 South Pacific tsunami. Earth-Science Reviews 107: 107–122.

CHAGUÉ-GOFF C, GOFF J, NICHOL SL, DUDLEY W, ZAWADZKI A, BENNETT JW, MOONEY SD, FIERRO D, HEIJNIS H, DOMINEY-HOWES D, COURTNEY C. 2012b. Multi proxy evidence for transpacific tsunamis in the Hawai'ian Islands. Marine Geology 299–302: 77–89.

CHAGUÉ-GOFF C, GOFF J, WONG HKY, CISTERNAS M. 2015. Insights from geochemistry and diatoms to characterize a tsunami's deposit and maximum inundation limit. Marine Geology 359: 22–34.

CHAGUÉ-GOFF C, CHAN JCH, GOFF J, GADD P. 2016a. Late Holocene record of environmental changes, cyclones and tsunamis in a coastal lake, Mangaia, Cook Islands. Island Arc 25: 333–349.

CHAGUÉ-GOFF C, SZCZUCIŃSKI W, SHINOZAKI T. 2017. Applications of geochemistry in tsunami research: a review. Earth-Science Reviews 165: 203–244.

CHAGUÉ-GOFF C, SUGAWARA D, GOTO K, GOFF J, DUDLEY W, GADD P. 2018. Geological evidence and sediment transport modeling for the 1946 and 1960 tsunamis in Shinmachi, Hilo, Hawaii. Sediment Geol 364:319–333.

CHAGUÉ-GOFF C, COPE J, KILROY C, JACOBSEN G, ZAWADZKI A, WONG H. 2020. A 7300-year record of environmental changes in a coastal wetland (Moawhitu), New Zealand, and evidence for catastrophic overwash. Sediment Geol. 407: 105746.

DONNELLY J, GOFF J, CHAGUÉ-GOFF C. 2017. A record of local storms and transpacific tsunamis, eastern Banks Peninsula, New Zealand. The Holocene 27: 496–508. DOI: 10.1177/0959683616670222

FERREIRA SLC, ANJOS JP, FELIX CSA, JUNIOR MMS, PALACIO E, CERDA V. 2019. Speciation analysis of antimony in environmental samples employing atomic fluorescence spectrometry – Review. TrAC Trends in Analytical Chemistry 110: 335–343.

IDRIS N, GONDAL MA, LAHNA K, RAMLI M, SARI AM, ALDAKHEEL RK, MITAPHONNA R, DASTAGEER MA, KURIHARA K, KURNIAWAN KH, ALMESSERIE MA. 2022. Geochemistry Study of Soil Affected Catastrophically by Tsunami Disaster Triggered by 2004 Indian Ocean Earthquake Using a Fourth Harmonics (λ = 266 nm) Nd:YAG Laser-in-

duced Breakdown Spectroscopy. Arabian Journal of Chemistry [103847].

IDRIS N, LAHNA K, FADHLI M, RAMLI M. 2017. Study on emission spectral lines of iron, Fe in laser-induced breakdown spectroscopy (libs) on soil samples. Journal of Physics: Conference Series 846: 012020.

IDRIS N, RAMLI M, KHUMAENI A, KURIHARA K. 2018. Detection of salts in soil using transversely excited atmospheric (TEA) carbon dioxide (CO_2) laser-induced breakdown spectroscopy (LIBS) by the aid of a metal mesh. Journal of Physics: Conference Series 1011: 012055.

KAYODE JS, YUSUP Y, NAWAWI MNM, ARIFFIN KS, KALIL AE, TAGWA MG. 2018. EDX-SEM-XRF data from selected Precambrian Basement Complex rock samples in part of Southwestern Nigeria. Data In Brief. 20: 1525–1531.

KELSEY HM, ENGELHART SE, PILARCZYK JE, HORTON BP, RUBIN CM, DARYONO MR, ISMAIL N, HAWKES AD, BERNHARDT CE, CAHILL N. 2015. Accommodation Space, Relative Sea Level, and the Archiving of Paleo-earthquakes along Subduction Zones. Geological Society of America.

KIKKAWA HS, NAGANUMA K, KUMISAKA K, SUGITA R. 2019. Semi-automated scanning electron microscopy energy dispersive X-ray spectrometry forensic analysis of soil samples. Forensic Science International 305: 109947.

MAY SM, VÖTT A, BRÜCKNER H, SMEDILE A. 2012. The gyra washover fan in the Lefkada Lagoon, NW Greece-possible evidence of the 365 AD Crete earthquake and tsunami. Earth, Planets, and Space 64: 859–874.

MITAPHONNA R, RAMLI M, ISMAIL N, KURIHARA K, SUBIANTO M, GONDAL MA, IDRIS N. 2021. Preliminary evaluation of chemical components in the 2004 Indian Ocean giant tsunami impacted soil using CO_2 laser-induced breakdown spectroscopy (libs). Journal of Physics: Conference Series 1816: 1012035.

MOORE A, NISHIMURA Y, GELFENBAUM G, KAMATAKI T, TRIYONO R. 2006. Sedimentary deposits of the 26 December 2004 tsunami on the northwest coast of Aceh, Indonesia. Earth Planets Space 58: 253.

MOREIRA S, COSTA PJM, ANDRADE C, PONTE LIRA C, FREITAS MC, OLIVEIRA MA, REICHART G. 2017. High resolution geochemical and grain-size analysis of the AD 1755 tsunami deposit: insights into the inland extent and inundation phases. Marine Geology 390: 94–105.

PARIS R, GOTO K, GOF J, YANAGISAWA H. 2020. Advance in the study of megatsunamis in the geological record. Earth Sci Rev 210: 103381. https://doi.org/10.1016/j.earscirev.103381

PARIS R, SABATIER P, BIGUENET M, BOUGOUIN A, ANDRE G, ROGER J. 2021. A tsunami deposit at Anse Meunier, Martinique Island: evidence of the 1755 CE Lisbon tsunami and implication for hazard assessment. Marine Geology 439: 106561.

PILARCZYK JE, SAWAI Y, NAMEGAYA Y, T. TAMURA, TANIGAWA K, MATSUMOTO D, SHINOZAKI T, FUJIWARA O, SHISHIKURA M, SHIMADA Y, DURA T, HORTON BP, PARNELL AC, VANE CH. 2021. A further source of Tokyo earthquakes and Pacific Ocean tsunamis. Nat Geosci 14: 796–800.

PUTRA PS, NISHIMURA Y, NAKAMURA Y, YULIANTO E. 2013. Sources and transportation modes of the 2011 Tohoku-Oki tsunami deposits on the central east Japan coast. Sedimentary Geology 294: 282–293.

PHILIP S, SINGH N. 2020. Comparative Soil Analysis by Scanning Electron Microscope: a Forensic Perspective. International Journal on Emerging Technologies 11(2): 915–923.

RAVANSARI R, WILSON SC, TIGHE M. 2020. Portable X-ray fluorescence for environmental assessment of soils: not just a point-and-shoot method. Environ. Int 134: 105250.

ROY K, SASADA K, KOHNO E. 2014. Salinity Status of the 2011 Tohoku-oki Tsunami Affected Agricultural Lands in Northeast Japan. International Soil and Water Conservation Research 2(2): 40–50.

RUBIN CM, HORTON BP, SIEH K, PILARCZYK JE, DALY P, ISMAIL N, PARNELL AC. 2017. Highly variable recurrence of tsunamis in the 7.400 years before the 2004 Indian Ocean tsunami. Nature Communications 10: 1038.

SEMBIRING H, GANI A, ISKANDAR T. 2008. Implications of salinity research in Aceh for Indonesian rice growing. International Workshop on Post Tsunami Soil Management, Bogor, Indonesia.

SHARMA R, PATEL KS, LATA L, MILOSH H. 2016. Characterization of urban soil using SEM-EDX. American Journal of Analytical Chemistry 7: 724–735.

SHINOZAKI T, SAWAI Y, HARA J, IKEHARA M, MATSUMOTO D, TANIGAWA K. 2016. Geochemical characteristics of deposits from the 2011 Tohoku-oki tsunami at Hasunuma, Kujukuri coastal plain, Japan. Island Arc 25: 350–368.

SIEH K, DALY P, EDWARD E M, PILLASZYK JE, CHIANG HW, HORTON B, RUBIN MR, SHEN CC, ISMAIL N, CHRISTOPHER VH, FEENER MR. 2015. Penultimate predecessors of the 2004 Indian Ocean tsunami in Aceh, Sumatra: stratigraphic, archeological, and historical evidence. J Geophys Res Solid Earth. p. 120.

SHINOZAKI T, GOTO K, FUJINO S, SUGAWARA D, CHIBA T. 2015. Erosion of a paleo-tsunami record by the 2011 Tohoku-oki tsunami along the southern Sendai Plain. Marine Geology 369: 127–136.

SHINOZAKI T. 2021. Geochemical approach in tsunami research: current knowledge and challenges. Geosci. Lett 8: 6.

SHINOZAKI T, SAWAI Y, IKEHARA M, MATSUMOTO D, SHIMADA Y, TANIGAWA K, TAMURA T. 2022. Identifying tsunami traces beyond sandy tsunami deposits using terrigenous biomarkers: a case study of the 2011 Tohoku-oki tsunami in a coastal pine forest, northern Japan. Progress in Earth and Planetary Science 9: 29.

SPISKE M, TANG H, BALHBURG H. 2020. Post-depositional alteration of onshore tsunami deposits – implications for the reconstruction of past events. Eath-Science Review 202: 103068.

SYUKRI M, ANDA ST, UMAR M, MEILIANDA E, SAAD R, FADHLI Z, SAFITRI R. 2022. Identification of tsunami deposit at Meulaboh, Aceh (Indonesia) using ground penetrating radar (GPR). International Journal of GEOMATE 23(96): 171–178.

SZCZUCIŃSKI W, NIEDZIELSKI P, KOZAK L, FRANKOWSKI M, ZIOŁA A, LORENC S. 2007. Effects of rainy season on mobilization of contaminants from tsunami deposits left in a coastal zone of Thailand by the 26 December 2004 tsunami. Environmental Geology 53: 253–264.

SZCZUCIŃSKI W, KOKOCIŃSKI M, RZESZEWSKI M, CHAGUÉ-GOFF C, CACHÃO M, GOTO K, SUGAWARA D. 2012. Sediment sources and sedimentation processes of 2011 Tohoku- oki tsunami deposits on the Sendai Plain, Japan—insights from diatoms, nannoliths, and grain size distribution. Sedimentary Geology 282: 40–56.

TROELS-SMITH J. 1955. Karakkteristering af lose Jordater (characterization of unconsolidated sediment). Denmarks Geologiske Undersogelse Series IV/3 10: 73.

VÖTT A, BRÜCKNER H, BROCKMÜLLER S, HANDL M, MAY SM, GAKI-PAPANASTASSIOU K, HERD R, LANG F, MAROUKIAN H, NELLE O, PAPANASTASSIOU D. 2009. Traces of Holocene tsunami across the Sound of Lefkada, NW Greece. Global and Planetary Change 66: 112–128.

WANG A, WANG Z, LIU J, XU N, LI H. 2021. The Sr/Ba ratio response to salinity in clastic sediments of the Yangtze River Delta. Chemical Geology 559: 119923.

WATANABE T, TSUCHIYA N, YAMASAKI S, SAWAI Y, HOSODA N, NARA FW, NAKAMURA T, KOMAI T. 2020. A geochemical approach for identifying marine incursions: implications for tsunami geology on the Pacific coast of northeast Japan. Applied Geochemistry 118.

WATANABE T, KAGAMI S, NIWA M. 2022. Geochemical and heavy mineral signatures of marine incursions by a paleotsunami on the Miyazaki plain along the Nankai–Suruga trough, the Pacific coast of southwest Japan. Marine Geology 444: 106704.

WILLERSHÄUSER T, VÖTT A, HADLER H, FISCHER P, RÖBKE B, NTAGERETZIS K, EMDE K, BRÜCKNER H. 2015. Geo-scientific evidence of tsunami impact in the Gulf of Kyparissia (western Peloponnese, Greece). Z Geomorphol 59(Suppl. 4): 43–80.

WIDIYANTO W, HSIAO S, CHEN W, SANTOSO PB, IMANANTA RT, LIAN W. 2020. Run-up, inundation, and sediment characteristics of the 22 December 2018 Sunda Strait tsunami, Indonesia. Nat Hazards Earth Syst Sci 20: 933–946.

XU J, GUO Y, YANG S, HOHL SV, ZHANG W. 2022. Reliable determination of SiO_2 concentrations in sediments *via* sequential leaching and ICP-OES/MS analysis. Journal of Geochemical Exploration 242: 107090.

YOSHII T, TANAKA S, MATSUYAMA M. 2018. Tsunami inundation, sediment transport, and deposition process of tsunami deposits on coastal lowlands inferred from the Tsunami Sand Transport Laboratory Experiment (TSTLE). Marine Geology 400: 107–118.

YU NT, YEN JY, CHEN WS, YEN IC, LIU JH. 2016. Geological records of western Pacific tsunamis in northern Taiwan: AD 1867 and earlier event deposits. Mar Geol 372: 1–16.